1.C51 final project report

Understanding solar panel adoption using the DeepSolar++ dataset

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1 Introduction

The objective of the project is to build on Prof Ram Rajagopal's work DeepSolar++: Understanding residential solar adoption trajectories with computer vision and technology diffusion models and to understand the factors that affect solar panel adoption in a given geographic location using various features. Once the best model is found, we can then interpret the model coefficients to help us understand what drives solar panel adoption. The dataset we will be using comes from the previous research of Rajagopal et al. and contains the number of buildings with solar panels for every year between 2005 and 2017 for every contiguous US census block [1, 2]. On top of that, the dataset contains some census block information such as median income, median age, average years of education, and global horizontal irradiation (GHI, a measure of solar power on the Earth's surface). To improve on their analysis, we will include a few other features which should be easy to get. For example, adding incentives per state or county, or results of political elections.

As our data is tabular, we will mostly be using simple regression models in our project. Those models are linear regression, neural networks, and some tree-based models such as random forest, gradient boosting, and XGBoost. We will compare each model by first performing hyperparameters tuning for each separately and then comparing their test scores. Interpretability of those model is straight forward except for neural networks. If the latter ends up being our most powerful model, we are planning on using SHAP values to help us understand which factors affect the solar panel adoption [3]. Figure 1 provides a summary of the steps we will go through to find the best model.

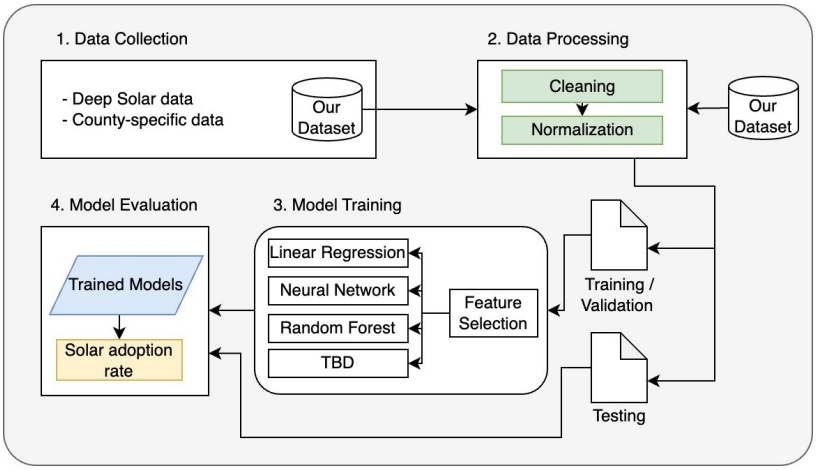


Figure 1: Schematic of the project steps.

Furthermore, by applying K-means clustering to the overall state features (by aggregating census-block data), it is possible to identify clusters of states that have similar patterns or characteristics in terms of features. This can help in identifying any potential outliers or anomalies in the data, which could have a significant impact on further analysis. Once a sufficiently powerful model is found, we will also apply the model to each state separately

to extract the most important features per state. Conducting the modeling on individual clusters will allow for a more focused analysis of the factors that contribute to solar panel adoption rates in each cluster and the barriers to adoption. By examining the characteristics of each cluster and identifying the key drivers of solar panel adoption within that cluster, it may be possible to develop more targeted policies and strategies to promote solar panel adoption in those regions.

Additionally, since we are dealing with time-series data (the number of residential PVs by year), we will also investigate time-series approaches to predict the cumulative number of residential PVs by 2017. This approach will focus on using both time-series data only, and combining time-series data with other tabular data we have in our dataset. This part is discussed in more detail in the time-series forecasting section.

2 Data

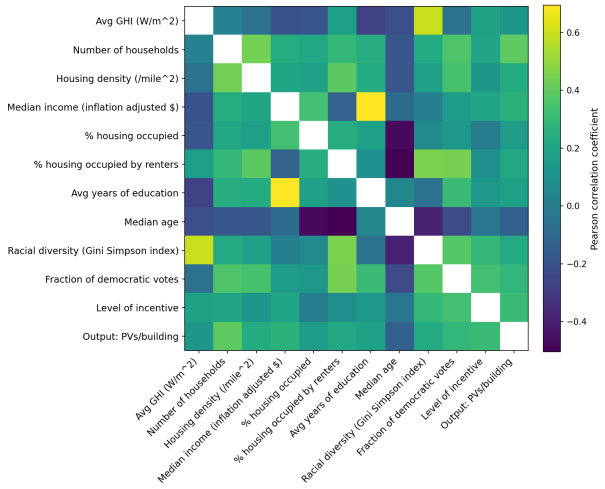
The data that we are using is broken down by census block, it contains 216 843 of them. Cleaning this to remove any rows that have missing values as well as removing census blocks where there were no building (and hence an infinite PV/building ratio), we are left with 208 613 census blocks. However, this level of granularity is too high for us especially as we are trying to find additional data sets to combine. Therefore, we aggregate the data by county leaving us with 3 106 counties in the end. We then perform some analysis on the correlation between our features. Figure 2a shows the full correlation matrix between each feature and between features and the output. We can see that the correlations are not very high, they are all between 0.1 and 0.4 in absolute terms. However, we can see that there are stronger correlations between our features. Especially the three pairs that are shown in figures 2b, 2c, and 2d.

2.1 Additional features

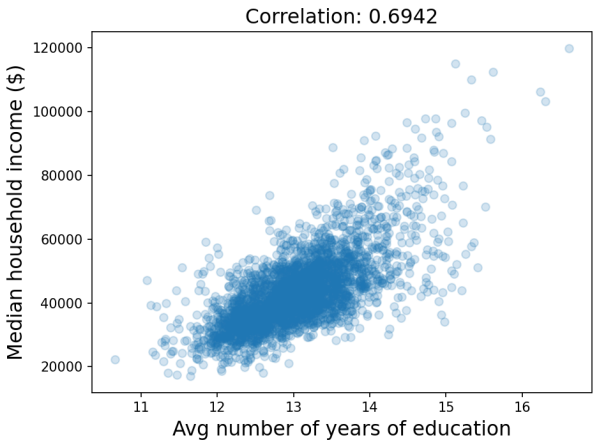
As part of the project, additional features were incorporated into the data set to enhance the models' performance in understanding and predicting solar panel adoption. State incentives for solar panels and partisan index (fraction of democratic voters) of a county were added as features.

The state incentives were obtained from the Energy Sage database and the Database of State Incentives for Renewables & Efficiency (DSIRE) from the North Carolina Clean Energy Technology Center[4]. State incentives may play a role in promoting solar panel adoption by providing financial incentives, tax credits, rebates, or other support mechanisms at the state level. By including state incentives as a feature in the analysis, the models can account for the impact of these incentives on solar adoption patterns.

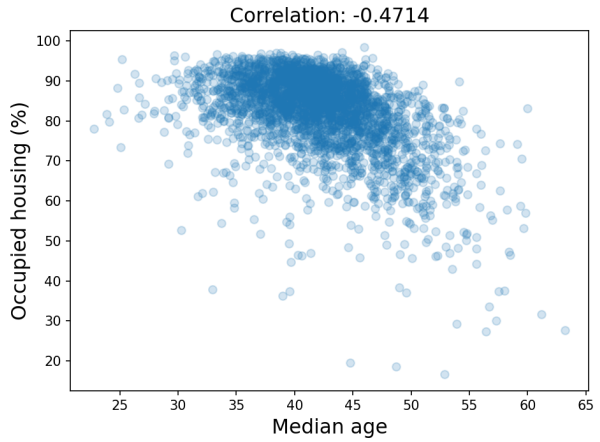
Based on the presence and extent of incentives for solar panel adoption, the states were classified into three categories: no state incentive, low state incentive, and high state incentives (see Figure 3a).



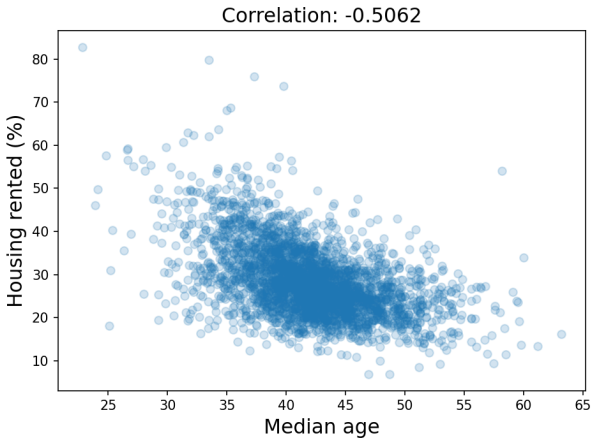
(a) Correlation with the output (PVs/building) and between features.



(b) Plot of median income vs years of education.



(c) Plot of median age vs percent of housing occupied.

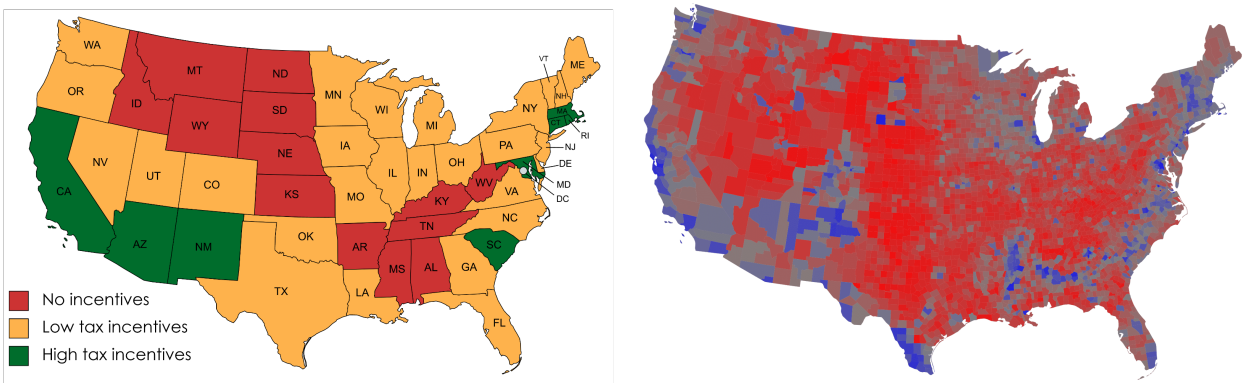


(d) Plot of median age vs percent of housing that is rented.

Figure 2: Various plots showing the correlations that exist between the features and output as well as between features. All the data shown is for 2016.

The states that had an absence of state incentives for solar panel adoption include Alabama, Arkansas, Missouri, and Wyoming. These states do not offer specific incentive programs or measures targeted at promoting the adoption of solar panels.

States that were classified to provide a low state incentive for solar panel adoption offer only one of the following measures: tax-free solar panel sale, net metering, low-interest loans for solar panel system purchases, or exemption from additional property tax due to addition of solar panel systems. These measures aim to provide some level of encouragement and support for residents and businesses interested in installing solar panels. Most states were classified to have low state incentives.



(a) (b)

Figure 3: (a) U.S. map classified by state incentives for solar panels. (b) County result of the 2016 elections, blue means a higher fraction of democratic votes and red a higher fraction of republican votes [5].

States were classified to provide high state incentives for solar panel adoption if they go beyond offering a single measure and provide a comprehensive range of incentives. These states recognize the importance of promoting renewable energy and offer multiple support mechanisms to encourage residents and businesses to adopt solar panels. High state incentives can include a combination of rebates, grants, and multiple tax incentives. California, New Mexico, Arizona, and South Carolina are examples of states that offer high incentives for solar panel installation and adoption.

Additionally, a partisan index (fraction of Democratic voters) feature was added on the county level (Fig. 3b). The values were based off the 2016 U.S. presidential elections. It allows for the exploration of whether there is a correlation between political preference and the likelihood of solar panel adoption.

2.2 State trends

K-means clustering was applied to the overall state features and then the results were plotted using dimensionality reduction, in order to gain insight into how different states compare to each other based on these features.

The District of Columbia (DC) was found to be an outlier. DC is a relatively small geographic area with a much lower population compared to many other US states. As a result, the overall state features used for clustering may be heavily influenced by the characteristics of the larger states (in terms of geography or population size), making it more difficult to accurately cluster DC within the same groups as the other states. Similarly, when DC was dropped and K-means clustering was applied again, the state of California was determined to be an outlier. due to its large population, size, and high adoption of PV. The resulting K-means clustering results in 3 clusters of states.

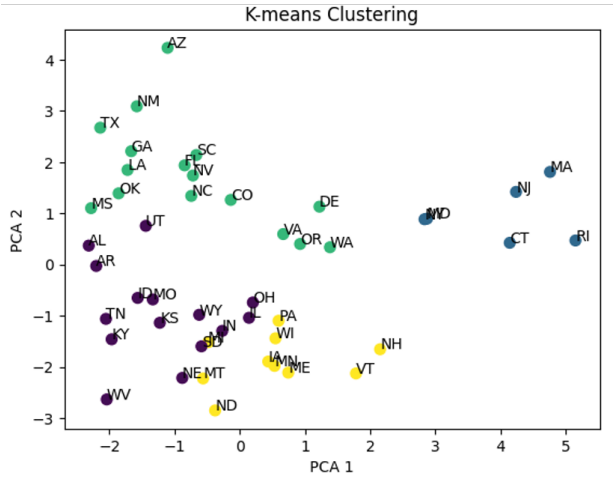


Figure 4: K-means clustering on features.

3 Best model for tabular data

3.1 Initial search

As the dataset comes subdivided by census block, we aggregated the data by county to attenuate the noise. The final features from the dataset used in the regressions are average solar irradiance, number of households, housing unit density, median household income, % housing units occupied, % housing units occupied by renters, average years of education, median age, racial diversity, and state of the county as a one hot encoded variable. All the metrics were for 2016. To this we added a partisan as well as incentive index from section 2.1. The variable we are predicting is the number of solar panels in 2016 divided by the number of buildings in a county. Counties without buildings were removed.

The first thing we did was to narrow down the list of potential models. This was done without the additional features. In table 1 the R^2 score of various models can be found. We performed 5-fold cross validation for each model on the training set (dividing it into training and validation) and selected the hyperparameter that lead to the highest average validation score. The best models of each type were then tested on the test set, and this

is the score that is reported in table 1. Firstly, we looked only at the features that were provided in the dataset. Models performed poorly with XGBoost being the best with an R^2 score of 0.3566. We then added state information as a one-hot encoding to the features. State information can be extracted using the census block FIPS code that was provided in the data. Models performed much better with RandomForest being the best with 0.7232 R^2 score. Since 38.80% of the data are counties without any solar panels, we also thought of looking at model performance using only counties with solar panels (where y>0). In that case, model performance improves again with XGBoost at the top with 0.8077 R^2 score.

The results in table 1 are expected. First of all, the models perform poorly without state feature because the conditions vary a lot from state to state. Including this information enables the models to take into account the differences between states and thus perform better. Furthermore, the model results when only taking into account counties with solar panels are mixed. Some models like GradientBoosting, XGBoost, and NN perform better while linear models perform much worse. The former would make sense as we are only splitting our problem in two and should, therefore, have more information to understand. The latter, however, could be explained by the smaller dataset or the fact that linear model learned something from the counties without solar panels.

Overall, linear models perform worse than other models. This is easily explained by the fact that our data must contain some non linear relationships that linear models cannot approximate. Furthermore, the poor performance of NNs might also indicate that our data is not easily separable. However, NNs performed very differently depending on the initialisation, more investigation would be needed to understand if they can perform better. Unsurprisingly, tree based models performed the best. Those models can extract complex relationships from the data and we can see this well here.

The better performance of models using only counties with adoption is interesting but explainable. Our task could be defined by a series of models, the first one predicting the presence or absence of solar panels in a county, and the second one predicting the amount of solar panel adoption. Intuitively, this split in our dataset could improve performance. However, after the hyperparameter tuning from section 3.2, the best models did not perform better with only y>0. The matter was dropped but it would be interesting to come back to it with a better look.

Table 1: R^2 score of various models and methodologies on the held out test set. The best model for each methodology is highlighted in bold.

	Lasso	Ridge	RandomForest	GradientBoosting	XGBoost	FNN
Without state features	0.1770	0.1796	0.3075	0.2124	**0.3566**	0.2806
With state feature	0.6550	0.6347	**0.7232**	0.6932	0.7079	0.5524
Only y>0	0.4963	0.4899	0.7106	0.7532	**0.8100**	0.7033

It is important to note that the scores reported in the table varied substantially with different train/test splits. To mitigate this, the split was stratified to give the same ratio of datapoints with y=0 and y>0 (except for the last methodology). Nonetheless, out of the different random seeds for data splitting, RandomForest and XGBoost were almost always the best. This hyperparameter search was simple and only included a single hyperparameter. This was the regularisation factor for Lasso and Ridge, and the number of estimators for tree based methods. The fully connected neural network was simply done by hand to understand its performance.

3.2 Hyperparameter tuning

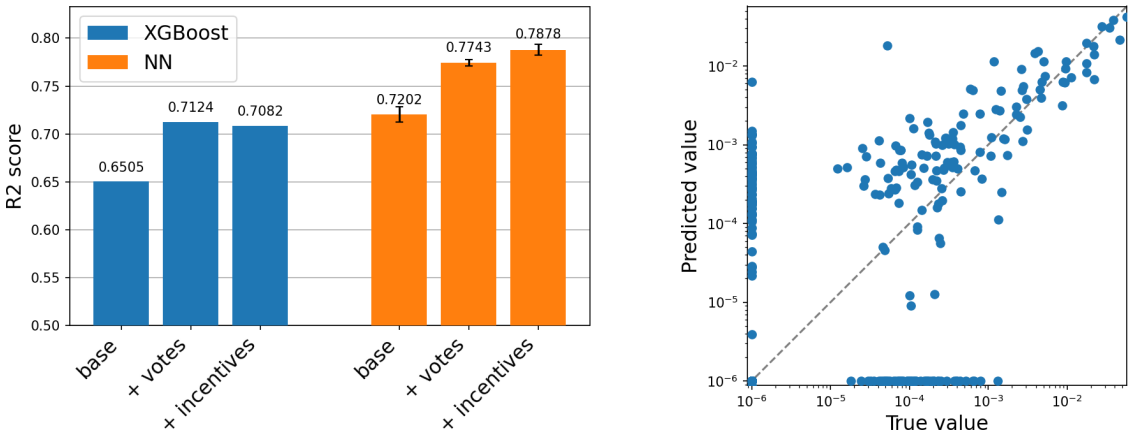
The result of the initial search showed that RandomForest and XGBoost were the most promising models to look at. We also decided to explore neural networks as not all architectures had been explored. For each model, a Bayesian optimisation was performed. At each iteration, the models were cross validated using the RepeatedStratifiedKFold function from sklearn. The number of folds was set at 5 and the number of repetitions at 10, thus totalling 50 cross validations with different sets. This was done to get a proper understanding of the mean and variance of model performance. Bayesian optimisation was chosen instead of random or grid search due to its efficiency in the number of iterations needed to find the best models. For this in depth tuning, the two extra features from section 2.1 were added and the performance with and without them was measured. It is important to note that the results in table 1 were found using a slightly different dataset than for this in-depth hyperparameter tuning. This is because a few counties and the "state" of DC were removed as they did not appear in either one of the two additional indexes. With a different number of rows, the train/test split was completely different even with the same random seed. The performance in this section is therefore slightly different and could appear worse when this is not the case.

During the tuning it was found that RandomForest was not improving as much as the other models, therefore it was dropped early on. For XGBoost, the parameters that were tuned were `n_estimators`, `max_leaves`, `learning_rate`, and `max_depth`. For the fully connected neural network, the number of nodes for the two layers, the initial learning rate, the activation function, and whether or not to add a final ReLU on the output were used as hyperparameters. The number of nodes for the first layer was varied from 20 to 100, and for the second layer, 0 to 50. The activation function could be any of ReLU, tanh, or sigmoid. A final ReLU was considered only if the output was not standardised as all of our data would be positive or 0. Finally, a StepLR scheduler was used for the learning rate as this is what provided the better results compared to a fixed rate or any other type of scheduling. Since the training of a neural network is different than that or a tree based method in that it requires multiple epochs, the bayesian loop was a little bit different. The various models were trained for a large number of epochs such that overfitting was sure to occur. The output of the training function for the optimisation was returned as the maximum value of the test set. This is because the training can be stopped earlier and any model that was found to be the best could have its number of epochs reduced to reach this peak.

Accounting for the fact that RandomForest was dropped early on and the additional features, a total of 6 Bayesian optimisations were run for the two models and 3 datasets. The results can be found in figure 5a. We can see that the additional features led to better test scores with the votes index leading to the highest jumpy. The results for the neural network were statistically significant and therefore, the last model, with 0.7878 R^2 score was chosen. This model had the following parameters:

- Number of nodes in layer 1: 80
- Number of nodes in layer 2: 0
- Activation function: tanh
- Learning rate: 0.012445653671345749
- Final ReLU: No

The parity plot for this model can be found in figure 5b.



(a) (b)

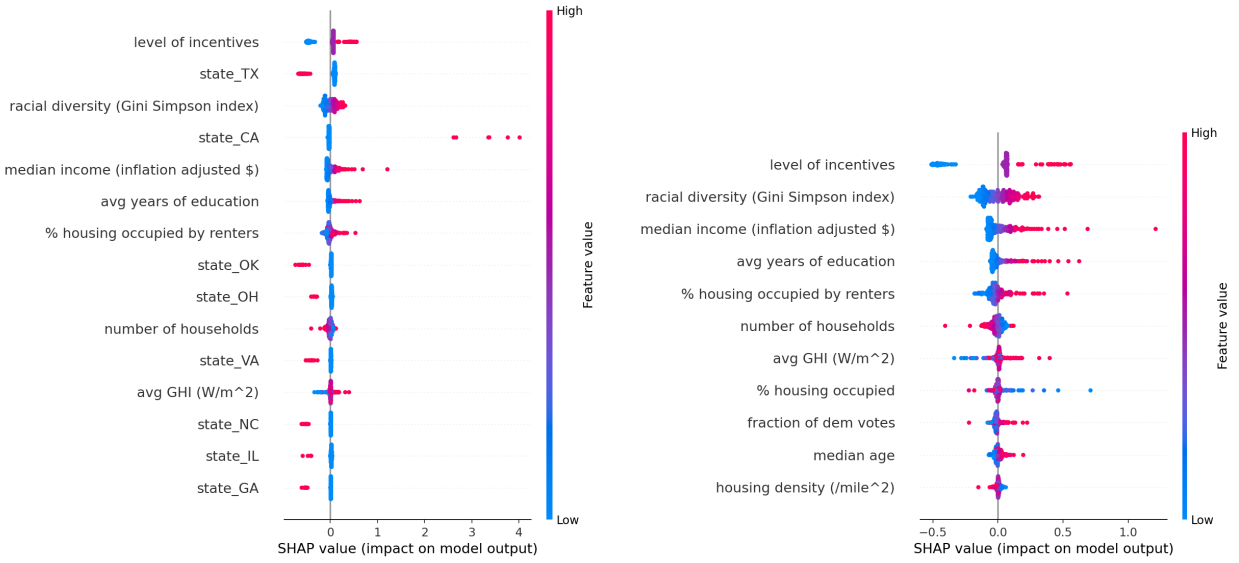
Figure 5: (a) Test set performance for the two types of models with different datasets. The error bars for the neural network indicate different initialisations. Note that the y axis does not start at 0 for ease of reading. (b) Parity plot for the best model found. For the log scaling, zero and negative values were were set to 10^{-6} which is a value smaller than the smallest non zero point.

4 Model interpretability

4.1 SHAP values

With the best model found, the task becomes to understand why it predicts the data the way it does and what we can learn from that. Neural networks are inherently very hard

to understand. To do so, we used SHAP values from the python package shap [3]. Figure 6 shows two beeswarm plots of SHAP values. In the first (fig. 6a) the top 15 features by absolute SHAP value are plotted. This figure contains the one-hot encoded state features. We see how important state features are as they make up 8 of the top 15 features. Interestingly, we can immediately see which states have a higher or lower adoption. California has a very strong positive effect on adoption while Texas, Oklahoma, Ohio, Virginia, North Carolina, Illinois, and Georgia has a strong negative effect. Removing those values and only looking at the 11 features that are not state ones we get the plot in figure 6b. As expected, the level of incentives plays a big role in determining solar panel adoption, the higher the incentive, the higher the adoption. Next, features such as median income, years of education, solar irradiance, and median age all show a positive effect as they increase. We would expect people with more money, education, that are older and live in sunnier areas to have more solar panels. Those findings align well with the paper that our study is based on. In their paper, Rajagopal et al. found that lower income income counties had a delayed adoption and saturated faster while incentives increased adoption especially in low and mid-income counties [1]. Additionally, we can see that the higher the percentage of democratic voters, the more the solar panels; albeit at a smaller absolute effect. Other features are harder to understand such as racial diversity, % of housing occupied, or housing density. Those might have a causal effect that is not clear with the current dataset.

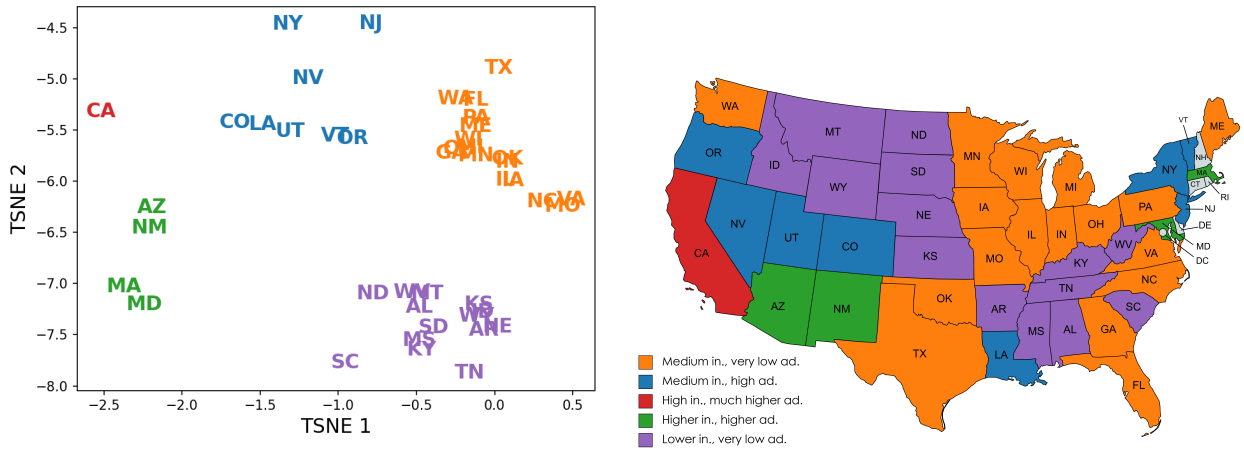


(a) (b)

Figure 6: SHAP values for different sets of features. (a) Using the top 15 sorted by absolute value, (b) the 11 features that are not state on-hot variables.

4.2 SHAP clustering

To further understand the drivers, we performed clustering on the mean absolute values of SHAP values for each state. For each state that had at least 20 counties, we extracted the SHAP values are average their absolute values. Each state then had a vector of length 11 (we removed the states features) on which we performed clustering. Figure 7a shows a TSNE plot of the clustering. We found the most stable clustering with a number of 5 clusters. More easily digestible, figure 7b shows the same clusters on a map of the conterminous United States. For each clusters we then looked at differences that were statistically significant between the counties in the states in the clusters and all other counties. We looked at this difference both in the dataset values to understand how those cluster differed, and the mean absolute values of SHAP values to understand how those clusters behaved. A summary of the results can be found in table 2.



(a) (b)

Figure 7: (a) TSNE plot of the states clustering based on SHAP values. (b) Corresponding US map coloured by clusters with legend on incentives and adoption.

Clusters 2 to 3 are very self explanatory. Cluster 4 are the states where there are very low incentives and correspondingly low adoption. The difference in factors compared to the rest of the states are that irradiance contributes less but incentives do. This makes sense, if those states have very low incentives, any amount will drive adoption higher faster. Clusters 2 and 3 both have high incentives and high adoption. The difference is how they respond to the factors found in the dataset. While California is motivated more by age, diversity, and income, Arizona, Maryland, Massachusetts, and New Mexico are motivated more by incentives and democratic votes.

The insights mostly come from observing the differences between clusters 1 and 2. Both sets of states have medium incentives but one has a high adoption and the other a very low adoption. In both clusters, incentives contribute to adoption less than for other states. What is important is that the democratic share of votes is a key difference here. In both sets

Table 2: Key differences in each cluster

	States	Feature differences	SHAP values differences
Cluster 0	CO, LA, NV, NJ, NY, OR, UT, VT	Medium incentives Higher adoption	Dem vote contributes more Incentives contribute less
Cluster 1	FL, GA, IL, IN, IA, ME, MI, MN, MO, NC, OH, OK, PA, TX, VA, WA, WI	Medium incentives Very low adoption	Income contributes less Dem vote contributes less Incentives contribute less
Cluster 2	CA	Higher incentives Very high adoption	Age contributes more Diversity contributes more Income contributes more
Cluster 3	AZ, MD, MA, NM	Higher incentives Higher adoption	Incentives contribute more Dem vote contributes more
Cluster 4	AL, AR, ID, KS, KY, MS, MT, NE, ND, SC, SD, TN, WV, WY	Lower incentives Very low adoption	GHI contribute less Incentives contribute more

the stronger the democratic lean, the higher the adoption, however, in counties from cluster 0, that relationship is stronger than in cluster 1.

5 Time-series forecasting

The goal of this project is to predict the adoption rate of PVs, and that is accomplished by dividing the number of PVs installed in a county over the number of houses. Therefore, estimating the number of PVs using historical data is the next logical step in this project. The goal of this section is to perform time-series forecasting on our data in order to predict the number of cumulative PVs installed in 2017, which will allow us to expand our PV adoption time-frame to later years. The time series forecasting will be accomplished using two approaches:

1. Statistical time-series forecasting approach. The methods we will cover in this preliminary results report are:
   (a) Auto Regressive Moving Average (ARMA)
   (b) Autoregressive Integrated Moving Average (ARIMA)
2. Deep learning approach

In this report our focus will be on the statistical time-series forecasting approach.

5.1 Statistical time-series forecasting

5.1.1 Step 1: Clustering using K-means and time series K-means

The first step in our approach is to first cluster our dataset and estimate a representative line for each cluster which will then we will predict the cumulative PVs installations for 2016 and 2017. Since we are dealing with a statistical approach, and no information can be learned by the models, we decided to focus on clustering the time-series data and finding representative lines rather than forecasting the installation for the whole dataset. We compared the performance of two clustering approaches, namely k-means and time-series k-means.

Elbow method and Inertia: We first started by using the Elbow Method to determine the optimal number of clusters for our dataset for the two clustering approaches. In our case the elbow was K = 6 for both k-means and time-series k-means (Fig. 8). Using 6 clusters, we calculated the inertia values achieved using each method, the k-means inertia value was equal to 21895250.98, where the time-series inertia value was equal to 132.90. This shows that the time series k-means was able to achieve drastically lower inertia value compared to k-means. Therefore, we decided to move on with this analysis using the time-series k-means results.

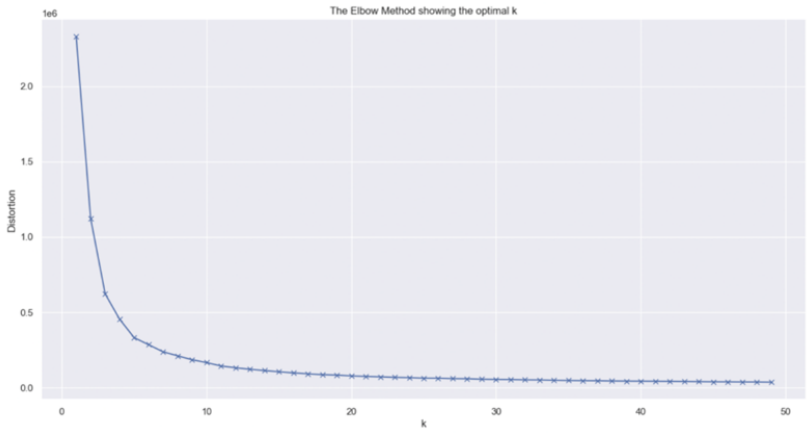


Figure 8: Elbow method to find the best number of clusters.

Finding a representative line for each cluster using time-series k-means: After identifying the optimal number of clusters and the inertia values achieved using k-means and time-series k-means, we then moved to finding a representative line for each cluster that captures the general trend seen in these clusters using the centroids of the time-series k-means clusters, which will then be used in our time-series forecasting step (Fig. 9).

5.1.2 Step 2: Time-series forecasting using ARMA and ARIMA

In our statistical time-series forecasting approach we used two methods, AMRA and AMIRA.

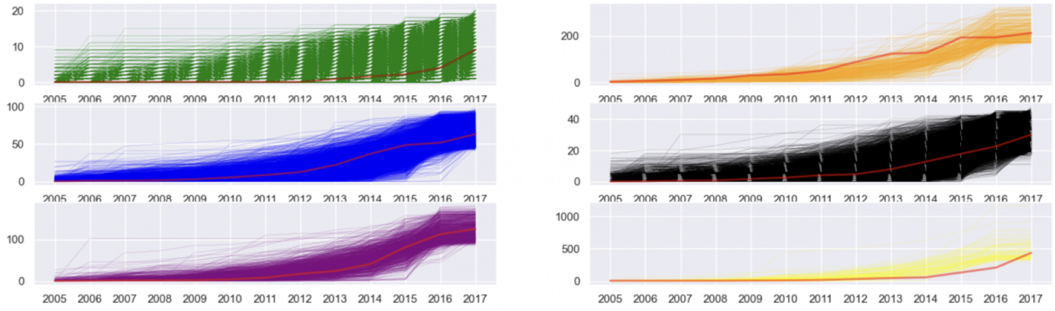


Figure 9: The six clusters of time series in our dataset with theit representatives.

- ARMA: The ARMA model is based on the concept of autoregression, which assumes that the future values of a time series variable can be predicted based on its past values. It also incorporates the idea of moving average, which considers the residual errors or the differences between the actual values and the predicted values.
- ARIMA: The main difference between ARMA and ARIMA is the I (Integrated component). This component models the difference of the time series variable to make it stationary.

Figure 10 shows the forecasting results we got using these two models, and in order to evaluate the performance, we used mean squared error as a metric. ARMA was able to achieve an MSE=8881.61, where ARIMA was able to achieve an MSE= 52233.41. Both results are relatively poor, and further adjustment for the two models' parameters is needed. Therefore, we will investigate using deep learning approaches to improve the results.

5.2 Deep learning time-series forecasting

5.2.1 RNN – LSTM

The method we used in deep learning time-series forecasting is bidirectional Long Short-Term Memory Networks (Bi-LSTM). LSTM is a special type of Recurrent Neural Network which is capable of handling the vanishing gradient problem faced by RNN. LSTM was designed by Hochreiter and Schmidhuber that resolves the problem caused by traditional RNNs and machine learning algorithms. Bidirectional LSTM (bi-lstm) is the process of making any neural network o have the sequence information in both directions backwards (future to past) or forward(past to future). Bi-LSTM usually perform better compared to LSTM, for this reason, we decided to use them in our project.

In this approach, we used to datasets, the first only consist of timeseries data of the number of PV installed, and the second combines that with the other features mentioned in the data section (not including the additional features). The training and testing results we got is shown in figure 11

Similar to the statistical approach, our goal is to forecast the cumulative number of installed PVs in 2017. Working with time-series data only, the model was able to achieve

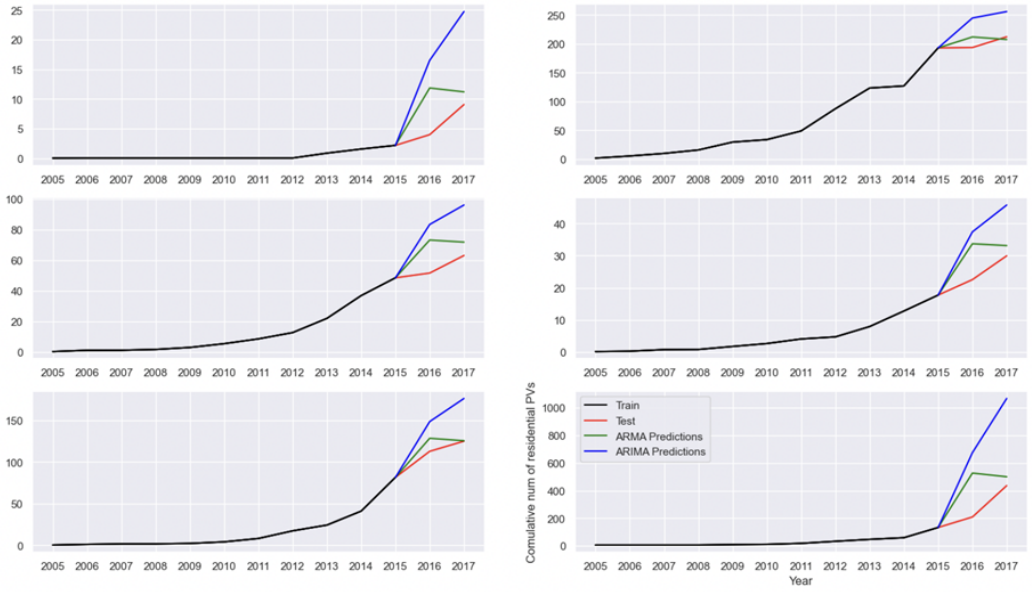


Figure 10: Results from ARMA and ARIMA on each of the six clusters.

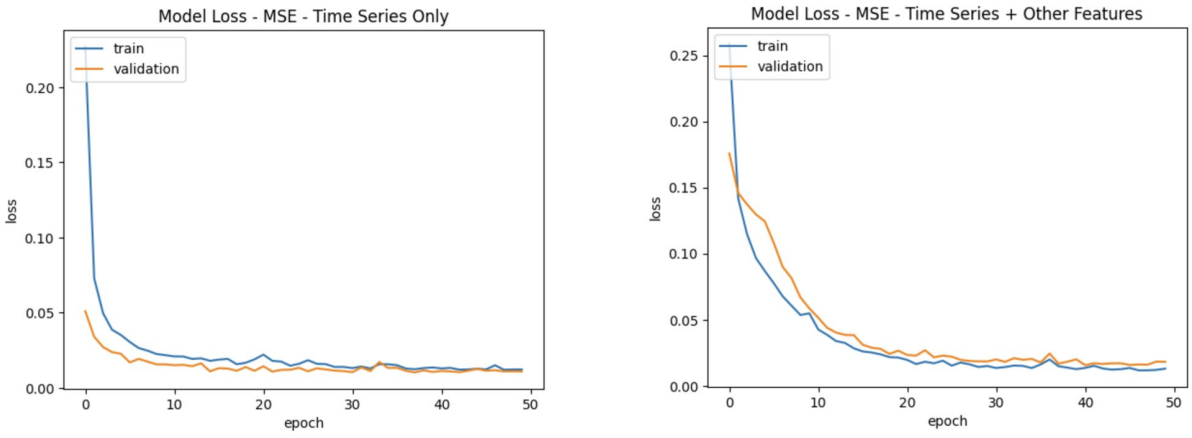


Figure 11: Training and testing losses with an LSTM model for time-series forecasting.

the performance shown in figure 12. The loss function used to train the network is MSE. Compared to the methods used for tabular data (section 3), the Bi-LSTM model was able to achieve higher r-squared results. However, it was interesting to see that the model was able to achieve slightly higher results using the timeseries data only compered to time-series+additional features. Understanding this results requires further investigation as part of our future work.

Dataset	Training loss (50 epochs)	Validation loss	Testing loss	R2 on test data
Time series only	0.0123	0.0109	0.01082	0.98983
Time series + additional features	0.0130	0.0182	0.01225	0.98810

Figure 12: Training loss, testing loss, and R-square with an LSTM model for time-series forecasting using two datasets.

6 Conclusion

In this project, we have looked at the DeepSolar++ dataset to understand the drivers to solar panel adoption. To the dataset, two new features were added, democratic lean and incentive level. The addition of those new features helped a fully connected neural network model to achieve an R^2 score of 0.7878 on the test set. Using this neural network, we were able to understand how different states behave using the current features. We found that a majority of states behaved as expected (lower incentives leading to lower adoption, and vice versa), but a minority provided important insights. Two clusters of states both had medium incentives, were not very affected by the level incentives and yet had very different adoption rates. The key difference seemed to be in the share of democratic votes where the positive correlation between democratic tendencies and adoption was stronger in one cluster than the other.

Those two clusters seem to be very important to study. While this current report only took a cursory look at the data, it would be interesting to dig deeper and understand the differences between those to sets of states to understand how to increase solar panel adoption in the long term. Furthermore, we have also shown that using an LSTM model improves our results substantially. While we only looked at tabular data to predict adoption for a single year, the LSTM results would suggest that incorporating the dimension of time that is present in the dataset would improve understanding of adoption. Finally, one last recommendation for future work would be to perform a deeper analysis on the splitting of the dataset between counties that have solar panel adoption and those that do not. This could help us understand, in a two step process, what are the key factors that drive the existence of solar panels, and what are the ones that increase solar panel spread once they have been introduced.

References

1. Wang, Z., Arlt, M.-L., Zanocco, C., Majumdar, A. & Rajagopal, R. DeepSolar++: Understanding residential solar adoption trajectories with computer vision and technology diffusion models. *Joule* **6,** 2611–2625. ISSN: 2542-4351. https://www.sciencedirect.com/science/article/pii/S2542435122004779 (2022).

2. Wang, Z., Arlt, M.-L., Zanocco, C., Majumdar, A. & Rajagopal, R. *DeepSolar++: Understanding Residential Solar Adoption Trajectories with Computer Vision and Technology Diffusion Model* https://github.com/wangzhecheng/DeepSolar_timelapse (2023).

3. Lundberg, S. M. & Lee, S.-I. in *Advances in Neural Information Processing Systems 30* (eds Guyon, I. *et al.*) 4765–4774 (Curran Associates, Inc., 2017). http://papers.nips.cc/paper/7062-a-unified-approach-to-interpreting-model-predictions.pdf.

4. *Database of State Incentives for Renewables & Efficiency - North Carolina Clean Energy* Database entry. https://www.dsireusa.org.

5. Weru, L. *Muddy America : Color Balancing The US Election Map - Infographic* https://stemlounge.com/muddy-america-color-balancing-trumps-election-map-infographic/?v=2 (2023).